

GRNT APEX

AMM Liquidity Handbook

Operating math for the binary CPMM, the 500m GRNT lender revolver, event lines, isolated margin, and why a pure vAMM is forbidden on a combat card.

This is the calculation book. Figures are a **model**. They are not live P&L, not a credit approval, and not a quote to a CPO. If a number in a deck conflicts with a formula here, the formula wins.

1. What this system is

APEX is a leveraged combat-sports terminal. Phase 1 is paper + CPA. The AMM is the Phase 3 cash engine, sold now so the CPO sees tase = 0. Players trade a **binary book** (YES/NO on a fight). The house is not the counterparty. GRNT is the lender of a short, liquid revolver. Ten brands on the same UFC main are **one book**.

Layer	Name	Job	Must not
L0	Senior facility	€500m revolving @ 6.5%, 7–30 day	Sit in one fight as k
L1	Junior first-loss	GRNT + operator. Halt lives here	Let senior pay winners
L2	Event line	Cap per fight / brand / group	Silo the same main 10 times
L3	CPMM book	$q_yes \cdot q_no = k$. Real cash layer	Set virtual L above the vault
L4	Isolated margin	Player posts m. Book sees m-lev	Cross-margin or portfolio net

Mode A vs Mode B

The curve does not change. **Mode A**: GRNT hosts the vault (operator is a broker). **Mode B**: the same CPMM sits inside the licensed casino vault; the 500m facility is a *hedge line*, not a sweep of player funds. Sell Mode B to MGA/UKGC. Flip a flag.

2. Notation

Symbol	Meaning	Base case
L	Event-line virtual+cash size ($q_yes + q_no$ at seed)	€10m = 2% of 500m
p	Oracle / seed probability of YES	0.62 favourite
q_yes, q_no	Pool reserves (cash units)	€3.8m / €6.2m
k	$q_yes \cdot q_no$, conserved on a fill	23.56×10^{12}
a	Cash in on a buy	€1.25m notional
s	Half-spread (base + inventory)	$2\% + 4\% \cdot \text{skew} $
N	Notional = margin $\times$ leverage	$\text{€}250\text{k} \times 5 = \text{€}1.25\text{m}$
F	Senior facility	€500m

3. Binary CPMM — cash layer

This is a **real complete-set AMM**, not a perp vAMM. Buying YES with cash a: mint a YES and a NO (complete set = €1 of collateral), then sell the a NO into the pool. k is unchanged.

Seed at oracle p

```

q_yes = L · (1 - p)
q_no = L · p
k = q_yes · q_no = L2 · p · (1 - p)
mid p_yes = q_no / (q_yes + q_no) = p
odds_yes = 1 / p_yes

```

Buy YES with cash a

```

q_no' = q_no + a
q_yes' = k / q_no'
YES from swap = q_yes - q_yes'
YES out = a + YES from swap
p_exec = a / YES_out
p_quote = min(0.99, p_exec · (1 + s))
odds = 1 / p_quote

```

Buy NO is the same with the reserves flipped. After a YES buy, q_{no} rises and q_{yes} falls: mid p_{yes} increases (favourite shortens). That is the inventory signal.

Worked fill — 10m line, $p = 0.62$, $a = €1.25m$

Step	Value
q_{yes}, q_{no} at seed	3 800 000 € / 6 200 000 €
k	23.560×10^{12}
$q_{no}' = 6.2m + 1.25m$	7 450 000 €
$q_{yes}' = k / q_{no}'$	3 162 416 €
YES out	1 887 584 € ($a + \text{swap } 637\,584\,€$)
p_{exec}	0.6622 (mid was 0.62)
impact $p_{exec}/p - 1$	6.8%
p_{quote} at 2% spread	0.6755 → odds 1.480

Lesson: €1.25m notional on a €10m line is a whale. Isolated 5x on €250k margin looks small to the player and large to the book. Cap notional against L, not against the player's deposit.

4. Depth — cash that moves mid by i

For a target mid p^* after a YES buy (no spread):

```

p* = q_no'^2 / (k + q_no'^2)
q_no' = √( p* · k / (1 - p*) )
a(i) = q_no' - q_no
with p* = min(0.99, p · (1 + i))

```

On the base book, cash to move mid +2% is **166 428 €**. That is the number to put next to “max 2% impact” on an event line. The facility-level shortcut $R \approx q \cdot (1-s)/s$ is the same identity for a one-sided peak q against a dedicated reserve.

Inventory spread and funding

```

s = s0 + k_s · |skew|
skew = (q_no - q_yes) / (q_yes + q_no)
funding_8h = f · skew (f = 8 bps at |skew| = 1)

```

Do not use 8-hour funding as a solvency tool. A fight is over in minutes. Funding is a skew tax between events, not a hedge during the round. Combat binary is a worse vAMM than a BTC perp: no time to pull inventory, settlement is 0 or 1, recreational flow sits on the favourite.

5. Allocation — never one 500m book

The senior facility is a revolver. It is allocated as lines. The same UFC main on ten casino brands is one inventory (group cap), not ten siloed AMMs.

Cap	% of F	On €500m	Rule
Event line	2%	10 000 000 €	One fight, one L. Default seed.
Group / correlated	4%	20 000 000 €	Same main across brands.
Brand	15%	75 000 000 €	One operator's total live lines.
Peak util of F	72% target	—	Halt new lev if a line >80%.

Topology	Drawn on 10 brands × one main	Verdict
Siloed (10 books)	100 000 000 €	Forbidden. Multiplies the one-sided hole.
Netting hub (GRNT)	20 000 000 €	Required. Inventory nets. Carry falls.
Saved	80 000 000 €	This is the point of the GRNT pool.

6. Facility sizing — ten large sportsbooks

How large a pool ten large casinos need is not “a month of handle”. It is peak one-sided flow at a max impact, plus a buffer.

```
monthly = n · H
weekly = monthly / (365/7/12)
q_peak = weekly · peak_share
R = q_peak · (1 - s) / s
need = R / target_util
```

Input / output	Base case
Casinos n	10
Sports handle H each	40 000 000 € / month
Combined monthly	400 000 000 €
Weekly	92 054 795 €
Peak one-sided (8% of weekly)	7 364 384 €
R at 2% impact	360 854 795 €
Need at 72% util	501 187 215 €
Facility F	500 000 000 € (covers need)
Avg drawn 40%	200 000 000 €
Interest 6.5% on drawn	13 000 000 € / year
Commitment 0.75% undrawn	2 250 000 € / year
Total carry	15 250 000 € / year
LP take 0.70% of handle	33 600 000 € / year
Coverage LP / carry	2.20×

€500m is the right senior size if $H \approx €40\text{m/month}$ per brand and peak is $\sim 8\%$ of weekly. If H is $€80\text{m}+$ or peak is 15% without a wider spread, add junior — do not enlarge the 6.5% loan. Tenor is 7–30 day revolving. Combat settles in hours. This is not a five-year loan.

7. Isolated margin (L4)

margin $m = \text{notional} / \text{lev}$, $\text{lev} \in [1, 20]$. The book sees notional. Ruin is capped to m . Liquidation unwinds inventory into the CPMM; the house is not the taker. If the liquidation queue exceeds 30s or the mark gap exceeds 2%: reject new 10x+, auto-deleverage. Cross-margin is forbidden. A parlay is a packaged book on the terminal, not a pooled margin.

Liquidation (approx.) for a YES at odds o , lev λ :

$\text{liq_odds} \approx o \cdot (1 - 1/\lambda)$ (isolated; haircut for spread/fees in production)

8. vAMM risk — why APEX is not Perp v1

A pure vAMM sets k virtually and holds cash in a vault. There is no natural other side. Open interest can be all YES. Winners are paid by an insurance fund. Perpetual Protocol abandoned this and rebuilt v2 on real Uniswap v3 liquidity. NFTPerp and Perp v1 (Luna/CREAM) are the case file. A 15-minute fight is a worse vAMM than a BTC perp: funding cannot mean-revert before the bell, settlement is binary, recreational flow is structurally long the favourite.

Risk	House	Pure vAMM	Real CPMM	APEX
Counterparty	The house	Fund / protocol	Other token in k	Cash nets; $\text{lev} \leq \text{junior}$
Depth	Liability cap	Fake. $k > \text{vault}$	Slippage = cash	$L \leq 2\% F$ and $L \leq \text{vault}$
Funding	—	Hours–days	Not for solvency	Do not use on a fight
Settlement	House pays	Must return to pin	YES=1 from cash	Official result + ADL
One-sided	Tase spike	Core death	Price moves, backed	Spread, halt, group cap
Oracle	Own traders	Mark $\oplus$ oracle	Pool is the price	Halt if lag $> 2s$
Liquidation	Lose the stake	Cascade / bad debt	No lev in base	Isolated; 30s / 2% halt
Backstop	Equity	Print token	Complete-set	Junior. Senior freeze

Favourite-win hole (pure vAMM, all YES)

```
margin = N / lev
payout = N · (1 – p)
uncovered = max( 0, payout – (vault – margin) – junior )
```

Input	Stressed example
Net YES notional N	8 000 000 €
Entry p	0.62
Leverage	5×
Vault / line	3 000 000 €
Junior	1 000 000 €
Margin posted	1 600 000 €
Winner payout $(1-p) \cdot N$	3 040 000 €
Uncovered	640 000 € — HALT. Cut N , cut lev, or add first-loss. Do not widen k .

APEX rule: complete-set cash for unlevered flow. Leverage is a **bounded** vAMM: max favourite-win payout $\leq$ junior. Senior never pays winners. Virtual L never exceeds the vault on that line.

9. Halt tape

Code	Trigger	Action
SKEW	One side >70% of a market book	Widen spread, hike funding, invite other side
UTIL	Inventory >80% of the allocated line	Halt new leverage on that event
NAV	Pool NAV –5% day or –8% event	Kill switch. Junior absorbs. Senior frozen
LIQ	Liquidation queue >30s or gap >2%	Auto-deleverage, reject new 10x+
ORACLE	Odds lag >2s vs feed	Quote only, no fill
CORR	Same-direction flow across ≥ 4 brands	Group cap. Treat as one book
TENOR	Position open >24h	Force funding or close — facility is short
CAPACITY	Need > $1.05 \times F$	Widen s, cut lev, or add junior
CARRY	LP take / carry < 1.0x	Raise skim or cut undrawn
VAMM	Uncovered hole > 0 on favourite win	Not a book. Reduce N or add first-loss

10. Operating rules (non-negotiable)

1. Never set virtual L above the vault on that line.
2. Never use funding as the solvency tool on a live fight.
3. Net the ten brands. Siloed vAMMs multiply the one-sided hole.
4. Cap OI so favourite-win payout $\leq$ junior. Senior does not pay winners.
5. Complete-set cash for unlevered flow. Leverage is bounded vAMM, not free k.
6. One event, one allocated line. Never one 500m Uniswap pool.
7. Mode A/B changes the legal host only. The curve is identical.
8. Phase 1 ships paper + CPA. Do not put DAZN, live AMM, or MGA Type 2 on the sprint.
9. Do not paste facility NPV into a first CPO mail. Hold mix is an assumption.
10. If any halt code trips, stop new leverage. Do not “wait and see” through a main event.

11. What the CPO is buying

A widget. Same 20x terminal. Dual-mode vault. GRNT senior revolver behind a junior first-loss. Tase 0% if the book is the pool and the house is not the taker. Commercial: SaaS from €6 000/mo + 15% of APEX-routed NGR. Revolut Metal is a Phase 2 on-ramp (50 € lock), not a Pay N Play rail. Trustly/Zimpler/Brite stay on the cashier.

13. AMM vs the exchange

A CPMM is a bonding curve, not a CLOB. Do not tell a CPO the fight book trades like a stock.

Kyle λ at seed (YES buy)

$$\lambda = dp/da \text{ at } a=0 = 2(1 - p) / L$$

On L = €10m, p = 0.62: $\lambda = 7.6 \times 10^{-5}$ per euro. Linear mid-move $\lambda \cdot Q$ on a €1.25m clip ≈ 0.095 . Exact CPMM mid-move is smaller (convexity). Always use the curve for fills, λ only as a risk tape.

Square-root law (Almgren, liquid equity)

$$\text{impact} \approx Y \cdot \sigma \cdot \sqrt{(Q / \text{ADV})}$$

$Y \approx 0.5$ (empirical; not a quote)

	Exchange / CLOB / futures	APEX CPMM
Mid	(bid+ask)/2, last, or index	$p = q_{\text{no}} / (q_{\text{yes}} + q_{\text{no}})$
Book	Discrete ticks, hidden size	Infinite infinitesimal bids on the curve
Impact	$\sqrt{Q}$. Path-dependent vs the tape	Closed form. $\sim$ linear in Q/L, then convex
Kyle λ	Estimated from flow. Unstable	Known, state-dependent. $2(1-p)/L$
Depth	Shares inside 1 / 5 / 10 bps	Cash a(i) that moves mid by i
Spread	Quoted + rebate. Avellaneda	$s_0 + \kappa \text{skew} $. Same idea, closed
VWAP	Vs the tape	$p_{\text{exec}} = a / \text{YES}_{\text{out}}$
Clearing	CCP, novation, variation margin	Complete set YES+NO = €1. Self-clears
Netting	Multilateral at the CCP	GRNT hub. Ten brands, one book
Margin	SPAN / portfolio. Cross-net	Isolated only. $m = N/\text{lev}$
Halt	LULD, volatility interruption	SKREW, UTIL, ORACLE, LIQ
Close	Auction, then T+1/T+2	Binary 0/1 at the bell. No unwind to pin
MM PnL	Spread – adverse selection	Spread + IL. No token to print

Exchange / CLOB / futures		APEX CPMM
Fake depth	Icebergs / spoof	vAMM k >> vault. Forbidden

Worked clip — €1.25m

Venue	Impact on €1.25m	Size that moves price 2%
APEX 10m line, p=0.62	Fill ~6.8%. Mid ~13%	~€0.17m
Nordic large cap, ADV €80m, $\sigma=2\%$, $Y=0.5$	~12.5 bps ($\sqrt{Q}$)	~€320m (not comparable)

Same euro clip is a rounding error on a cash equity and a whale on a fight line. That is why event cap is 2% of the 500m facility, not ADV of the sport. SPAN-style portfolio margin is forbidden: a parlay is a packaged book, not a net.

14. Source of truth

Code: src/lib/liquidity.ts, lender.ts, vamm-risk.ts, exchange.ts. Interactive: APEX app → Lender pool. Public read: github.com/isomarkkupentti/grnt-apex-playbook. If the app and this PDF diverge, fix the PDF in the same commit as the code.